

Attune — an AI radio station with the engagement dial turned the other way

Solo: concept → code → production → user study. BSc thesis artifact, ZHAW School of Management and Law · April–June 2026 · michael.moeckli@swissonline.ch

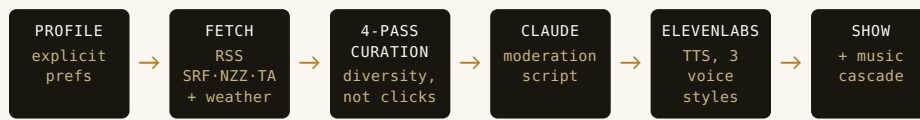
THE PROBLEM

Automated media runs on one stack — recommenders, LLMs, synthetic voices — tuned to one KPI: time-on-platform. The cost side (doomscrolling, filter bubbles, eroded trust) is well documented, and broadcasters and wellbeing products keep circling the same question: **can the identical technology be pointed at the listener's wellbeing instead, and still produce something people actually want to use?**

That was the question of my BSc thesis — but it is a product question, not a literature question, so I committed to answering it with a **working production system and real users**, not a mockup: a fully automated, personalized radio where every algorithmic choice optimizes autonomy, diversity and a defined session end instead of retention.

WHAT I BUILT

Attune generates a complete radio show on demand — greeting, curated news, weather, music, an audible ending. No feed, no autoplay, no account: explicit preferences in, finished show out.



Next.js 15 / React 19 / TypeScript on Vercel serverless · music via Spotify → Jamendo → local fallback · mock mode for every external service

2 days

FIRST COMMIT →
FIRST END-TO-
END SHOW

9 weeks

ZERO →
PRODUCTION →
VALIDATED

64

COMMITTS, SOLO,
BESIDE FULL
COURSE LOAD

~4.8k

LINES OF
TYPESCRIPT

Decisions I'd defend in a review:

- **An inverted recommender.** Four deterministic curation passes — *coverage* → *serendipity* → *depth* → *fallback* — optimize topic diversity and time-of-day fit instead of engagement. The selection rationale is exposed in the UI: every listener can see why a story made their show.
- **Prompt engineering as a discipline, not vibes.** Few-shot GOOD/BAD pairs enforce source fidelity; a transition typology shapes segment handovers; mid-show context is passed between generation calls — which fixed the model cheerfully re-greeting listeners in segment three.
- **Guardrails in code.** A source check verifies the generated script only cites articles that were actually fetched. Mock modes let the whole pipeline run in demos and tests without burning API budget.
- **Serverless trade-offs made explicit.** Vercel functions mean a read-only filesystem and a 60-second cap — so audio travels as data URLs and shows generate segment-wise. Constraints shaped the architecture; I didn't fight the platform.

TEILNEHMER-ID

SCHRITT 2 / 4 · DEIN PROFIL

1 / 2-3 empfohlen für 10 Min

THEMEN

Politik International Schweiz Wirtschaft Finanzen
Sport Technologie **Kultur** Wissenschaft Gesellschaft

Zurück

Zunächst erhält jedes gewählte Thema eine Meldung. Wenn noch Platz ist, kommt eine Überraschung ausserhalb deiner Auswahl dazu, danach folgt eine zweite Meldung pro Thema.

STANDORT

Zürich

MODERATIONSTYLL

Seriös Wie SRF Locker Wie ein Privatsender Energetisch Mit viel Begeisterung

SENDERLÄNGE

5 Min Kurz 10 Min Standard 15 Min Ausführlich

MUSIK

Mit Musik 2 Songs Ohne Nur Sprache

Ohne Musik dauert die Sendung etwa 3 Min. Die Songs verlängern sie um rund 6 Min — gesamt also ca. 9 Min.

Sendung vorbereiten · ca. 9 Min

The profile step (German UI): explicit topic, style, length and music choices — with a plain-language explanation of how curation will use them, and a footer naming every external service. Nothing is inferred behind the listener's back.

WHO USED IT — AND HOW FEEDBACK CHANGED THE PRODUCT

Apr 4–6

First commit to first working end-to-end show in two days. Demo-able from week one; everything after was iteration on a running system.

May

Formative test, 3 guided sessions (full SUS: mean 79.2). The sharpest feedback wasn't a feature request: the working title "Radio 25" made testers benchmark it against FM radio and confuse it with a local station. I renamed and repositioned the product — **Attune** — and iterated the UI before the pilot.

May 19 — Jun 1

Two-week pilot in production behind an invite gate: **25 participants** generated and listened to their own shows on their own devices, unsupervised, then answered a structured survey (usability, wellbeing constructs, free text; n = 25 after consent filter).

WHAT CHANGED — MEASURED

82.5/100

ADAPTED USABILITY INDEX (SUS-BASED)

4 / 4

RESEARCH SUB-QUESTIONS ABOVE SCALE MIDPOINT

14 / 25

WANT TO KEEP USING IT AFTER ONE SESSION



Mean ratings, scale 1–5 · gold line = scale midpoint (3) · n = 25

The answer the project needed: **explicit preference setting plus transparent curation is operable by ordinary users and measurably supports perceived autonomy** — same stack as the attention economy, inverted target, validated with real listeners rather than asserted.

WHAT I LEARNED (THE HONEST PART)

- **Transparency has a salience problem.** Users rated the transparency features helpful — when they noticed them. Several never saw the per-story rationale at all. Showing your work isn't enough; the UI has to interrupt with it at the right moment. That finding reshaped the thesis discussion more than any positive score.
- **My guardrail missed a failure class.** One pilot show claimed a football club had "won yesterday" — they hadn't played. The source check verified *which* articles were cited, not *temporal claims about them*. I designed a recency/claims guard in response, and took the broader lesson: evals have to target the failure you haven't seen yet, not the one you already fixed.
- **Split adoption is a roadmap, not a verdict.** The 11 of 25 who wouldn't continue named two concrete causes in free text — general AI skepticism and the lack of a native app — both addressable, neither fatal to the concept.

HOW I BUILD

Daily collaboration with AI tooling: Claude Code on feature branches, PRs reviewed against my own bar before merging; hi-fi UI states AI-designed, then implemented by hand; the survey analysis is a reproducible Jupyter notebook — every figure regenerates from the raw CSV. All AI use is declared in the thesis. Working transparently and fast with these tools is part of the craft, not a footnote.

Nine weeks, one person: a fuzzy problem turned into a production system, put in front of 25 real users, and answered with numbers — including the unflattering ones. That loop — sit with users, ship, measure, iterate — is the job I'm applying for.